C H A P T E R 1

Getting Started

This chapter will help you unpack your Macintosh PiForma, set up its traveling companions, and start the computer. You will also learn how to check a reluctant battery, shut down properly, and return everything to the bag. The packing order is worth a little attention. The foam has opinions.

Unpacking the equipment

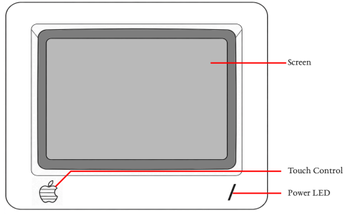
Set the carrying bag upright on a stable surface and open it. Leave room nearby for each item as you take it out.

1. Remove the small brown accessory pouch near the top. It contains two Nintendo 64 controllers, the USB hub, and the power cable. Set the pouch and its contents aside.
2. Take out the mouse and the collapsible computer stand beneath the pouch.
3. Remove this printed guide.
4. Lift out the mouse pad beneath the guide.
5. Remove the velvet-covered foam board beneath the mouse pad. Keep it flat; it protects the computer when everything is packed.

CHAPTER 1

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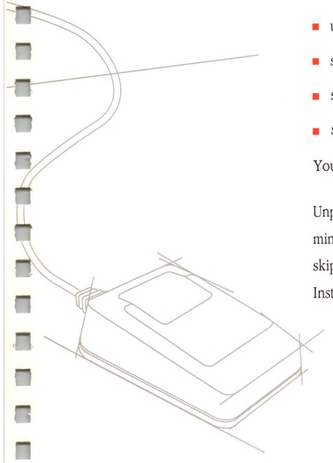
Getting Started

THIS CHAPTER WILL HELP YOU GET STARTED WITH YOUR MACINTOSH® PLUS computer. You'll find instructions for

- unpacking the equipment
- setting up the Macintosh Plus
- starting the computer
- starting the tour disk, *Your Apple Tour of the Macintosh Plus*

You'll also find important safety instructions.

Unpacking and setting up your Macintosh Plus is easy and takes only a few minutes. If somebody has already set up your computer for you, you can skip most of this chapter; go directly to the "Important Safety Instructions" section and continue on from there.



The power switch does not disconnect external charging power from the internal charger. Switching the computer off alone does not make internal electrical work safe. Do not disassemble it with charging power connected. Internal repairs and battery disconnection belong with someone familiar with this particular build, not in a first-start procedure.

Stop using the computer if it becomes unusually hot, smells hot, or has damaged power wiring or a damaged battery. Do not keep retrying power-on in that condition. Use the intended charging equipment. This privately built device has no claimed electrical-safety certification.

Starting up the computer

1. Find the physical power control on the right side and switch the PiForma on.
2. Look for the green light at the front.
3. Listen for a startup chime. Depending on the rear selector's position, there may be no sound.
4. Watch for the startup picture or sequence. The desktop normally follows in about one to two minutes; allow for some variation.

A silent startup does not by itself mean that the computer has failed. Use the green light and display as well as the chime when checking what has happened.

Checking the battery

If nothing happens, check the battery before assuming something has broken. The outside plunger mechanically presses the button on the internal battery pack, showing its charge indicators; the owner reports that pressing it may also wake or enable the pack.

The normal, management-approved way to read the battery is to press the rear battery plunger to turn on its display, then peer through the slits in the top to see the percentage. You do not have to open the compartment for this check.

The startup chime

A silent startup does not by itself mean that the computer has failed. Use the green light and display as well as the chime when checking what has happened.

Starting up the computer

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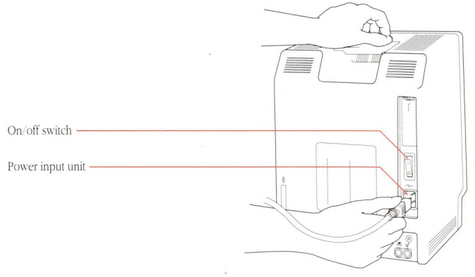
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The power cord

- **Plug the socket end of the power cord into the power input unit on the back of the Macintosh Plus. It's below the on/off switch.**



- **Plug the other end of the power cord into a three-hole, grounded outlet.**

△ **Important** Don't turn on the computer just yet. Wait until you've connected all the parts of your system and you've read the "Important Safety Instructions" section later in this chapter. △

The PiForma is underneath the board, with its monitor/screen facing upward. The keyboard is tucked against one side of the computer. It is not another layer in the stack. Watch that it does not catch as you remove the PiForma.

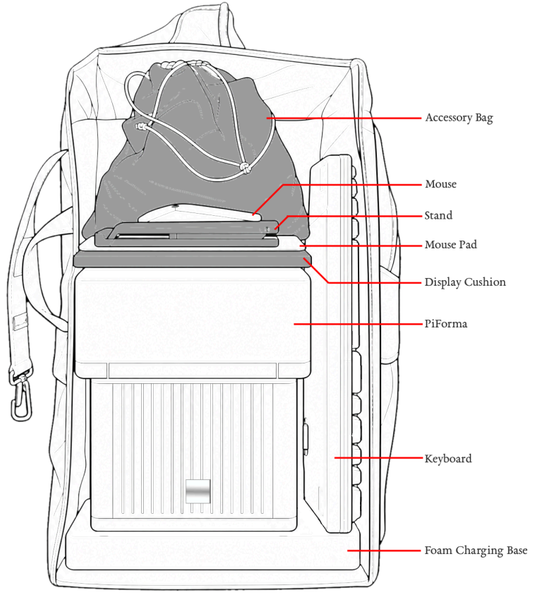


Figure 1-1. Arrangement in the carrying bag. The printed guide is not shown; it lies beneath the mouse and stand, above the mouse pad.

Unpacking the equipment

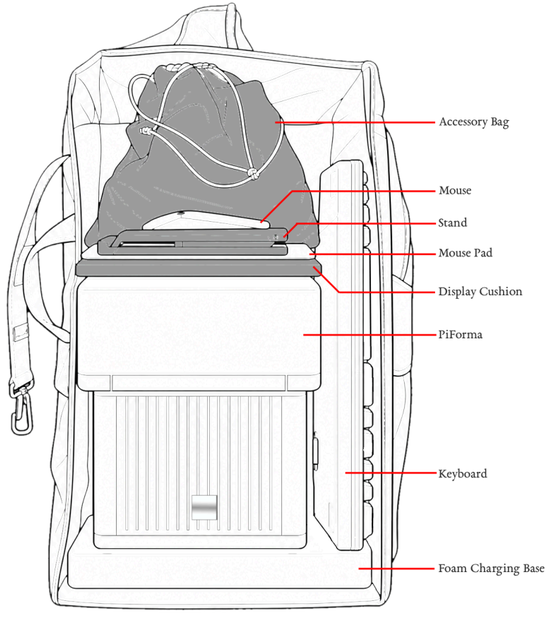
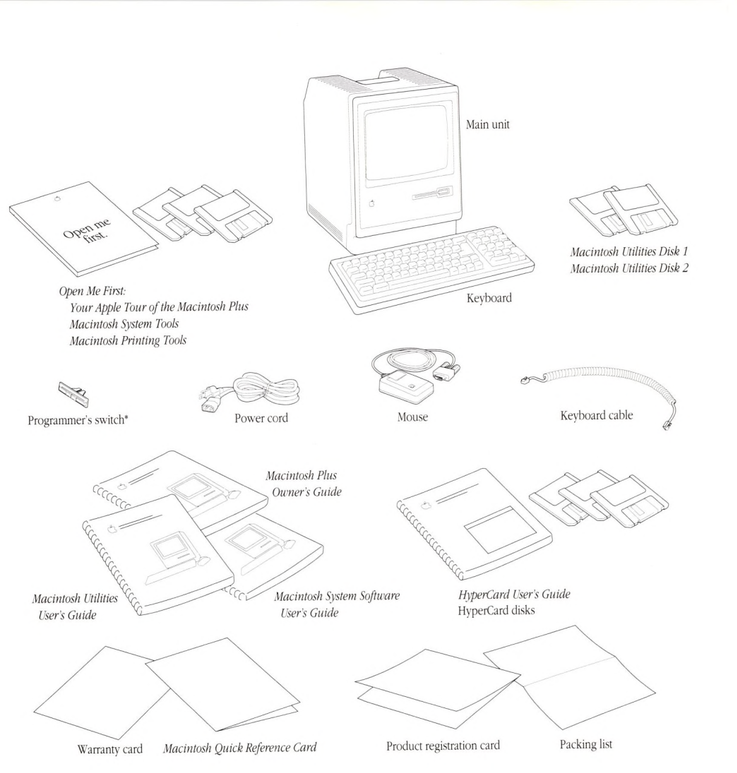


Figure 1-1. Arrangement in the carrying bag. The printed guide is not shown; it lies beneath the mouse and stand, above the mouse pad.



* Put this aside for now. You won't install the switch unless you're developing application programs for the Macintosh Plus.

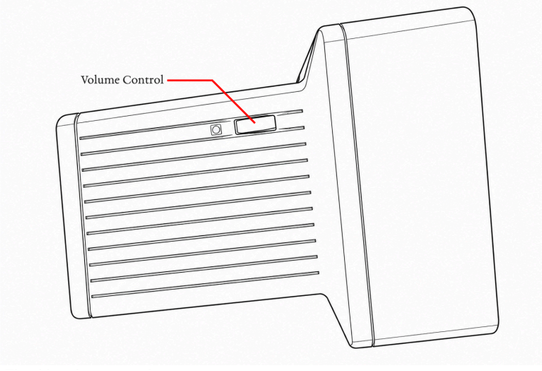


Figure 1-3d. The side volume control.

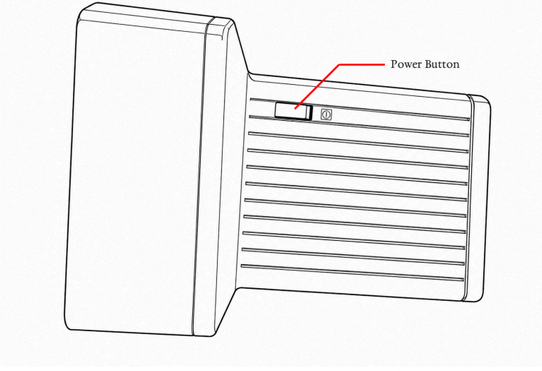


Figure 1-3e. The physical power button on the opposite side. This is separate from the Apple-logo touch control.

Taking a closer look

Before connecting anything, find the power control on the right side, the green status light at the front, and the chime-selector dial at the back. The Apple logo at the front is also a touch control. It is separate from the physical power switch.

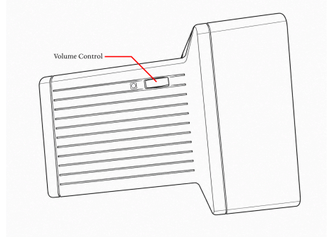


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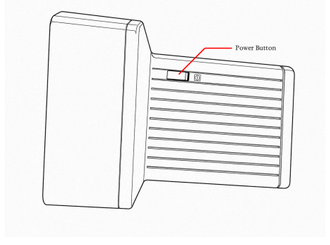
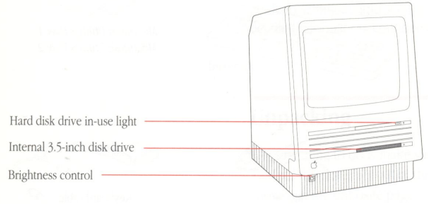


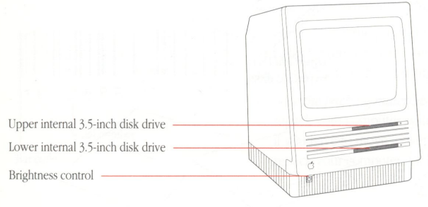
Figure 1-3e. The physical power button on the opposite side. This is separate from the Apple-logo touch control.

Taking a closer look

Now that you have your Macintosh SE out of the box, take a closer look so you're familiar with its parts.



Macintosh SE with internal hard disk drive



Macintosh SE with second internal 3.5-inch disk drive

Everything you connect to the Macintosh SE attaches to connectors, or **ports,** on the back panel of the computer. You can usually tell which port to use by looking at the **icons** (small pictures) above each one. You probably won't need them all right away; you'll learn more about some of them in the next few sections. (All the ports on the back panel are discussed in more detail in Chapter 4, "Macintosh SE Ports and Buses.")



Figure 3-1. View → Show Desktop returns you to the launcher without quitting the open applications.

Working with application windows

Several programs can be open at once. Click a window to bring it forward and direct your typing to it. Before typing a command or entering text, check that the intended window is active. An overlapping window is still open even when you can see only an edge of it.

Where a normal window has a title bar, drag it to a more useful position. Where a resize border or corner is provided, drag that border or corner to change the window's size. Some programs fix their window size or use the whole screen; those controls will not always be present.

Use a program's Quit or Exit command to leave it. A window close control closes that window, which may leave other windows from the same program running. A save question gives you a chance to keep changes; Cancel returns you to the work so you can decide.

Showing the desktop hides windows from view without saving or quitting them. Return to an open window by clicking its visible part or the appropriate Control Strip pane. Give the strip a moment to detect a newly opened program. Returning to a running program is different from deliberately opening another copy of it.

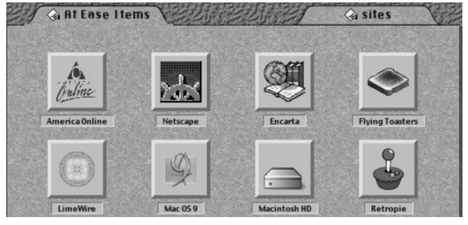
Window controls

The themed document window in Figure 3-2 has a close control at the upper left. At the upper right, the far-right control maximizes the window; the control immediately to its left minimizes it to the Dock, or Control Strip. Minimizing puts the window out of the way while leaving it open. Use the program's Control Strip pane to return to it.

Opening programs with AtEase

AtEase

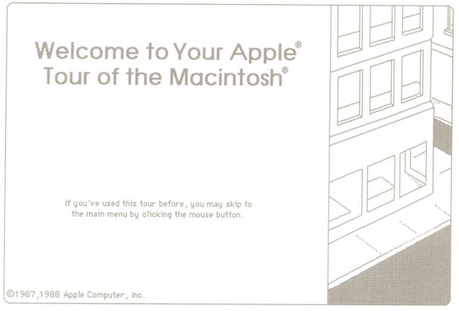
Click an AtEase folder tab to choose a collection, then click a program button once. Allow time for the program to open. A launcher button starts a program; it is not the document you may later create inside it.



Show Desktop

If application windows cover the launcher, choose View → Show Desktop from PiForma Panel. This reveals the desktop without quitting the programs. You can then select another AtEase button. If the launcher is absent rather than covered, see Appendix B.





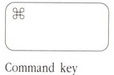
- **Follow the instructions that appear on the screen.**

Use the tour disk as long as you like. Anything you need to know to use the disk will be explained right on the screen.

Problems starting up?

If you don't see the opening display, go through the following list to see if you can identify the problem:

- Is the computer plugged into a power source?
- If your computer is plugged into a power strip, is the power strip turned on?
- Is the computer power switch turned on?
- Is the brightness control adjusted correctly?
- Are you using the correct disk? To check, hold down the Command and Shift keys while you press the 1 key on the main keyboard. This will eject the disk. (You'll learn more appropriate ways to eject disks later, but use this Command key shortcut for now.) If it is the correct disk, reinsert it into the disk drive.
- If you have a hard disk drive, and it's already been prepared for you, did you turn off the Macintosh Plus before inserting the disk?



Command key

If you can't identify the problem yourself, get help from a more experienced Macintosh user or from your authorized Apple dealer or representative.

Appendix A: Technical Information

This appendix summarizes the inspected project records. It is not a wiring plan or a certification. Compare identifying labels and the running configuration before using it as a current-device specification.

Hardware inventory and identification

Donor: Apple IIc Monitor A2M4090, recorded February 1985.

Computer: Raspberry Pi 4 Model B, recorded 4 GB RAM.

Storage: recorded 256 GB microSD.

Display: recorded 9-inch LCD kit, working mode 800 × 480.

Main battery: recorded Anker Prime 26K 300W.

Internal charger: recorded 100W USB-C adapter.

Keyboard: recorded Apple A9M0330 ADB keyboard, BLE conversion, ZMK, USB-C charging, nice!view.

Mouse: recorded Bambu kit electronics in a printed shell, USB receiver.

Startup sound: recorded YS-M3, CK1028 selector, separate salvaged speaker.

Normal sound: recorded internal USB speaker.

Recorded power architecture

Recorded path: rear AC inlet → internal charger → main battery → original power switch → custom 5V JST rail → Pi, LCD controller, startup-chime module, and touch sensor. The USB speaker is supplied through the Pi. The switch does not disconnect the AC charging path. The rail has no separate installed fuse protection in the records.

Technical Information

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	Macintosh (128K)	Macintosh 512K	Macintosh 512K Enhanced	Macintosh Plus	Macintosh SE	Macintosh II
Internal 3.5-inch disk drive	400K	400K	800K	800K	800K Optional second 800K	800K Optional second 800K
External 3.5-inch disk drive	400K	400K	400K or 800K	400K or 800K	400K or 800K	Not supported
Hard disk	None	Optional external Hard Disk 20	Optional external Hard Disk 20	Optional external Hard Disk 20 Optional external SCSI hard disks	Optional external Hard Disk 20 Optional SCSI hard disks (internal/ external)	Optional SCSI hard disks (internal/ external)
SCSI	None	None	None	SCSI port (DB-25)	SCSI port (DB-25)	SCSI port (DB-25)
Serial ports	Two serial ports (DB-9)	Two serial ports (DB-9)	Two serial ports (DB-9)	Two serial ports (Mini-8)	Two serial ports (Mini-8)	Two serial ports (Mini-8)
Video display	Built-in 9-inch diagonal 512-by-342 pixel black-and-white monitor	Built-in 9-inch diagonal 512-by-342 pixel black-and-white monitor	Built-in 9-inch diagonal 512-by-342 pixel black-and-white monitor	Built-in 9-inch diagonal 512-by-342 pixel black-and-white monitor	Built-in 9-inch diagonal 512-by-342 pixel black-and-white monitor. Accepts optional external monitor connected through video card in SE Bus expansion connector.	Supports multiple external color and monochrome monitors connected through video cards in NuBus expansion slots.
Keyboard	Macintosh keyboard (RJ-11)	Macintosh keyboard (RJ-11)	Macintosh keyboard (RJ-11)	Macintosh Plus keyboard—includes numeric keypad (RJ-11)	Apple Keyboard or Apple Extended Keyboard connected through Apple Desktop Bus™ ports (Mini-4)	Apple Keyboard or Apple Extended Keyboard connected through Apple Desktop Bus ports (Mini-4)
Mouse	Macintosh Mouse connected through mouse port (DB-9)	Macintosh Mouse connected through mouse port (DB-9)	Macintosh Mouse connected through mouse port (DB-9)	Macintosh Mouse connected through mouse port (DB-9)	Apple Desktop Bus Mouse (Mini-4)	Apple Desktop Bus Mouse (Mini-4)